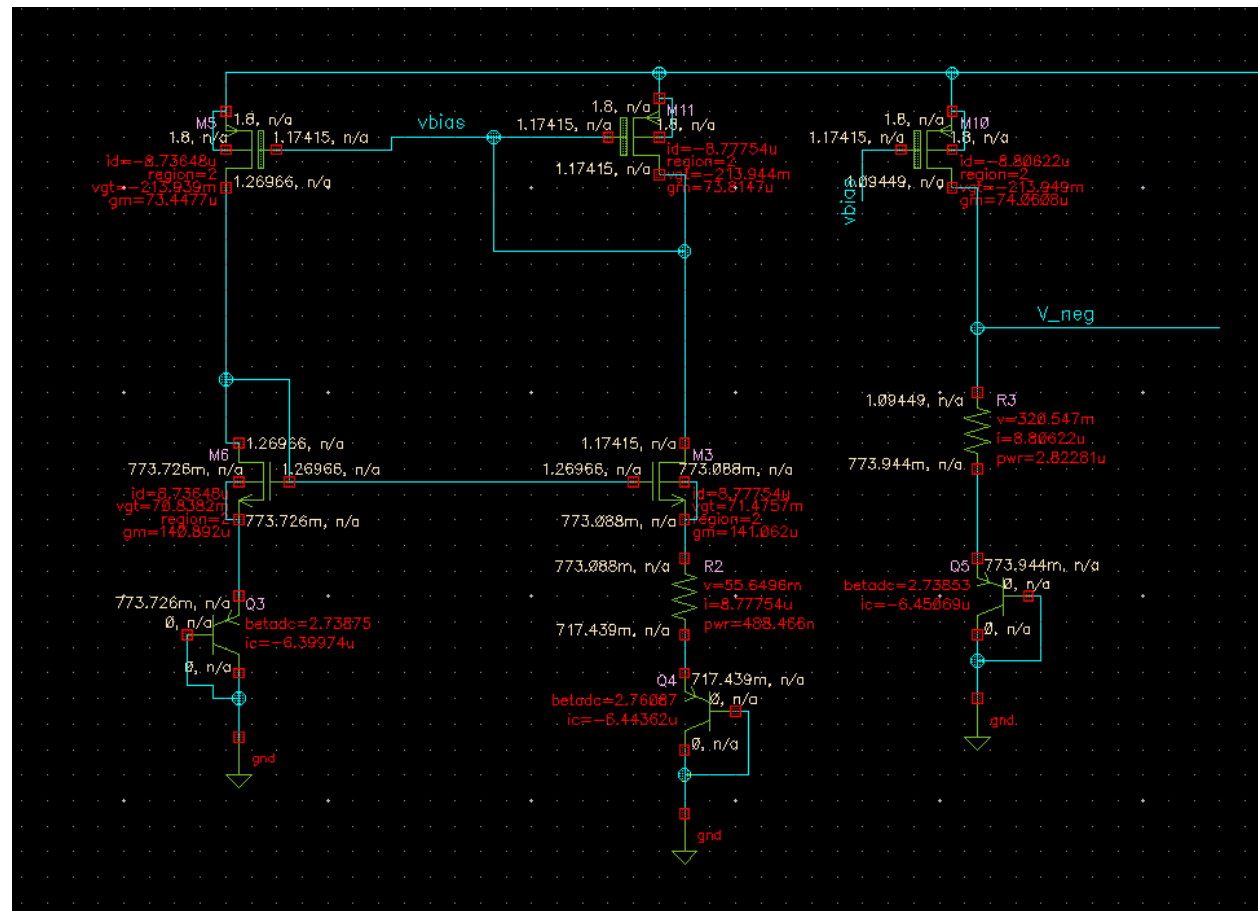


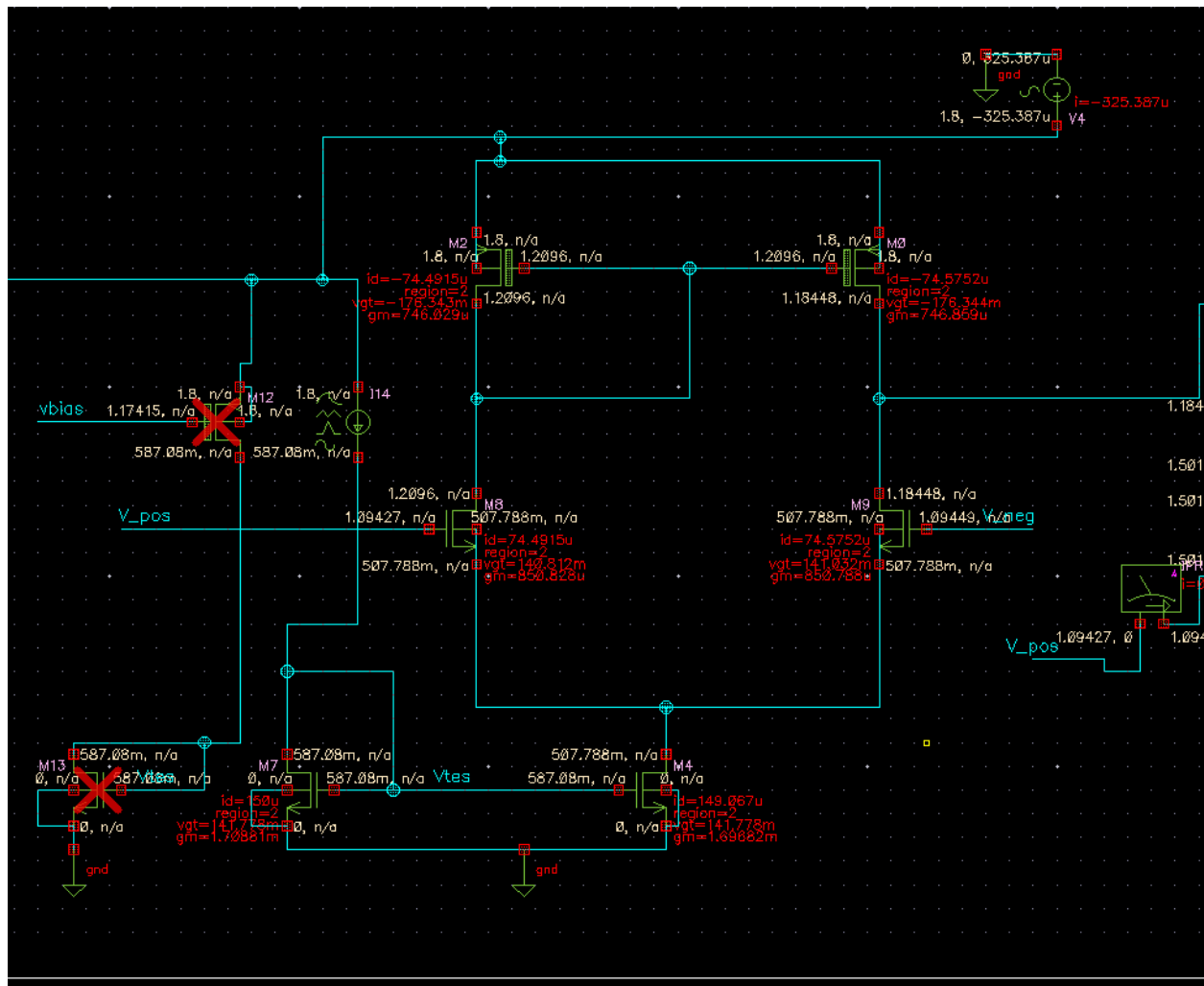
SAHASRITH BOOTLA (23110064)

1. Band Gap Reference



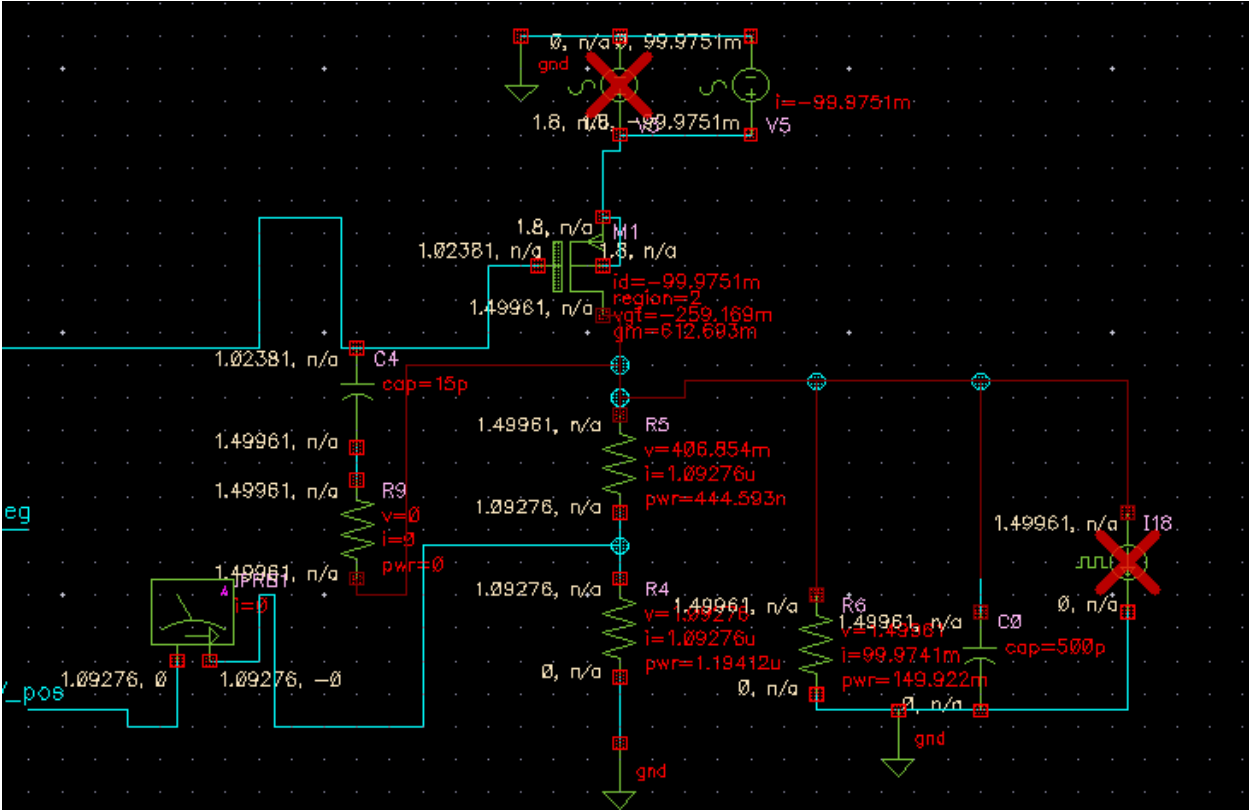
MOSFET	Type	$ V_{ov} $ (mV)	W (μm)	L (μm)
M5	PMOS	213.9	7	1
M6	NMOS	70.8	7	1
M11	PMOS	213.9	7	1
M10	PMOS	213.9	7	1
M3	NMOS	71.4	7	2

2. Operational Transconductance Amplifier



MOSFET	Type	Vov (mV)	W (μm)	L (μm)
M7	NMOS	141	25	0.54
M4	NMOS	141	25	0.54
M9	NMOS	141	12.5	0.54
M0	PMOS	176	84	1
M2	PMOS	176	84	1
M8	NMOS	139	12.5	0.54

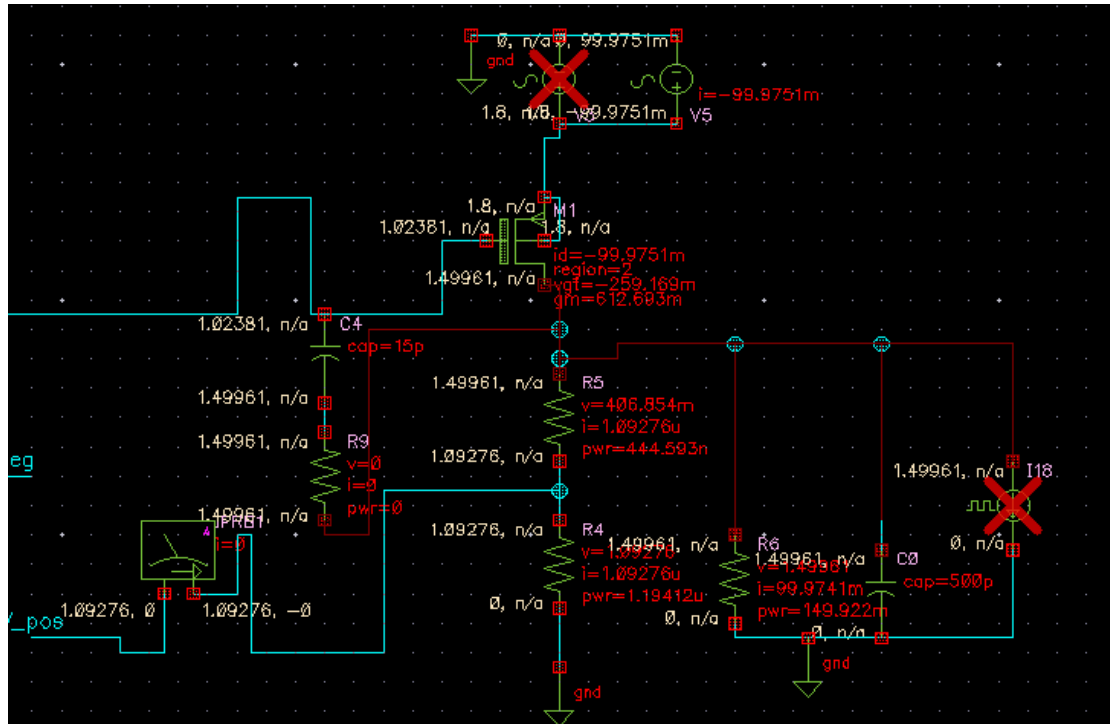
3. Low Dropout Regulator



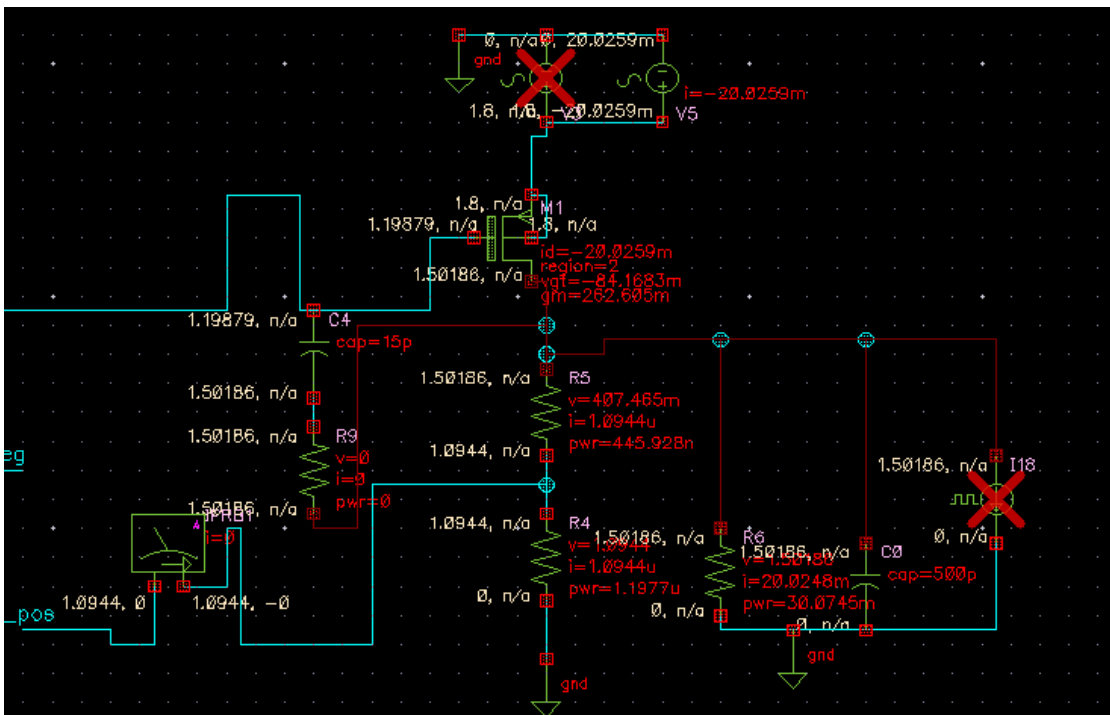
MOSFET	Type	Vth (mV)	W (μm)	L (μm)	Multiplier
M1	PMOS	259	50	0.18	120

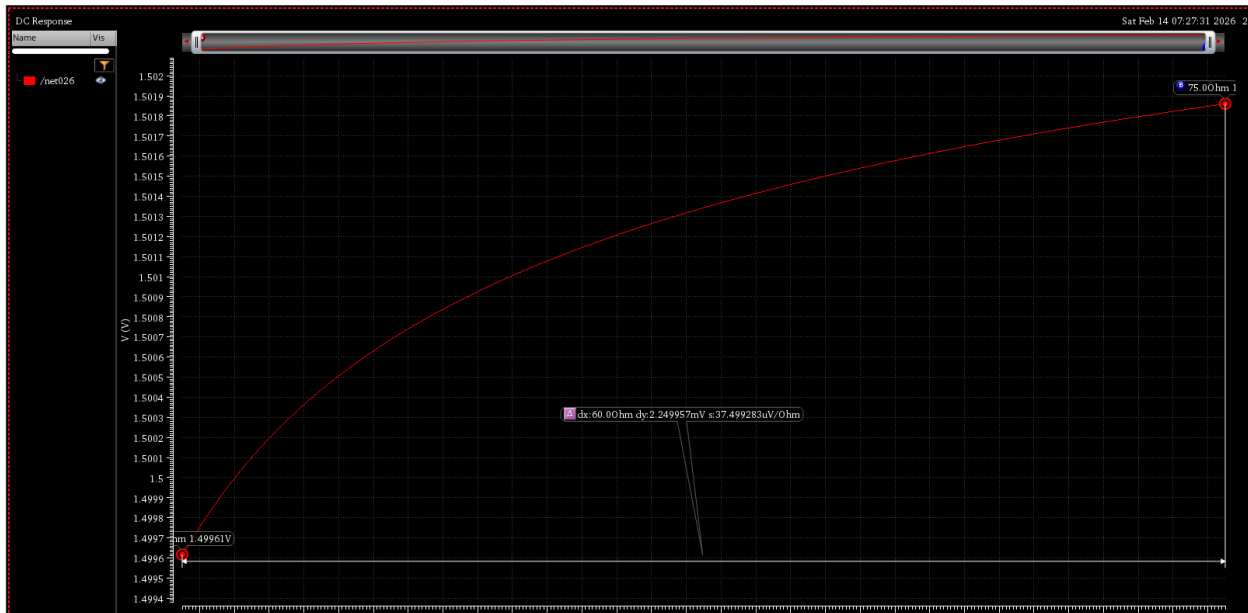
B. Load regulation : $\leq 2\%$ variation from light-load to full-load

1. High load



2. Light Load



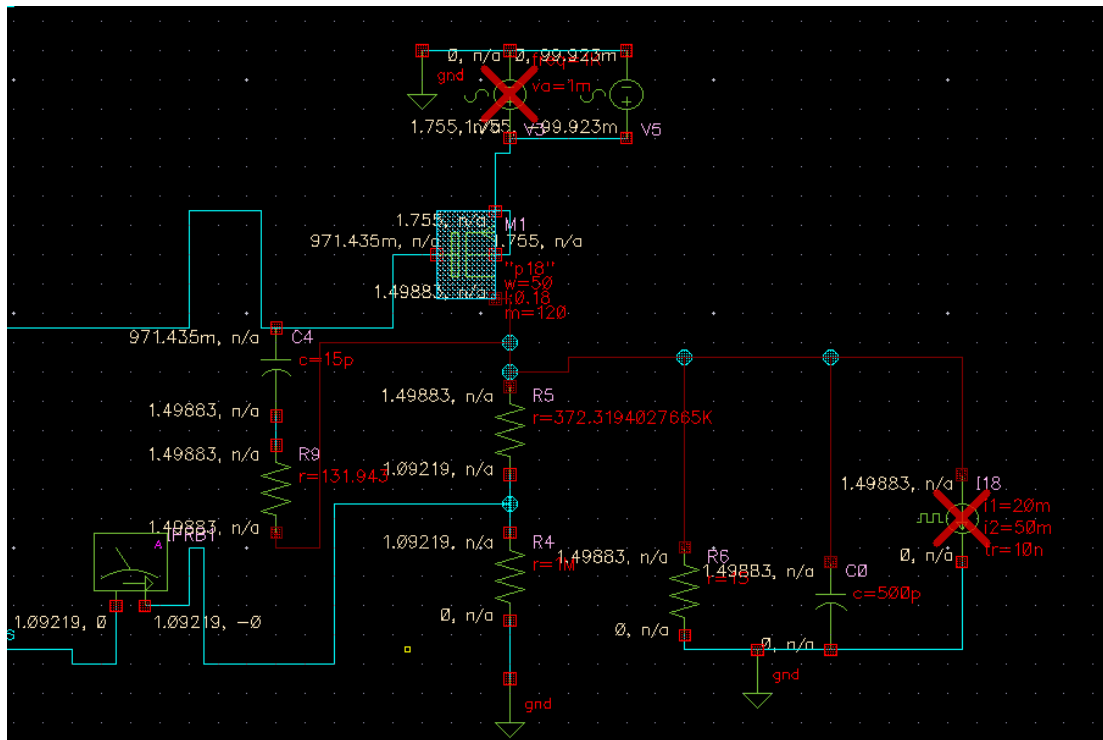


LOAD	Voltage(V)	Current(mA)
Full load	1.49961	99.9741
Light load	1.50186	20.0248

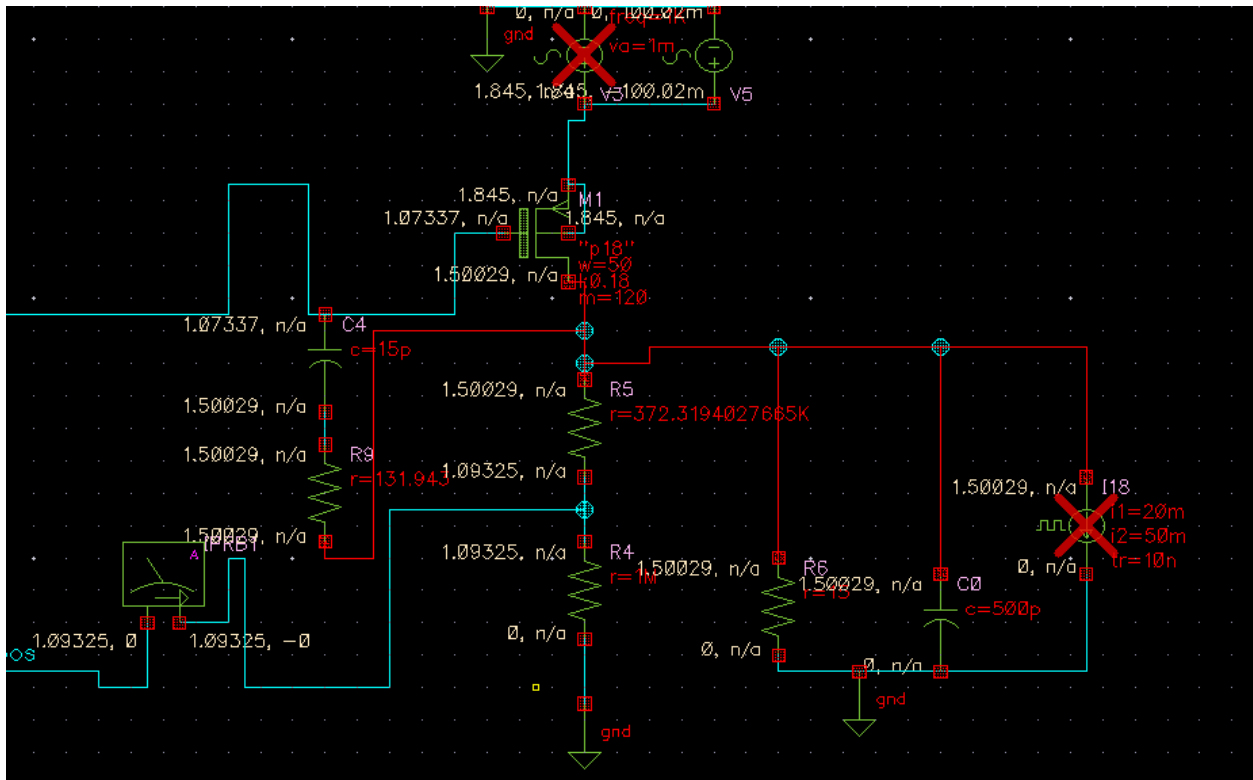
- Load Regulation $\approx 0.028 \text{ mV/mA}$

C. Line Regulation $\leq 2\%$ for $\pm 2.5\%$ variation in input voltage

1. $V_{\text{input}} = 1.8 - 0.045 = 1.755\text{V}$



2. $V_{input} = 1.8 + 0.045 = 1.845V$



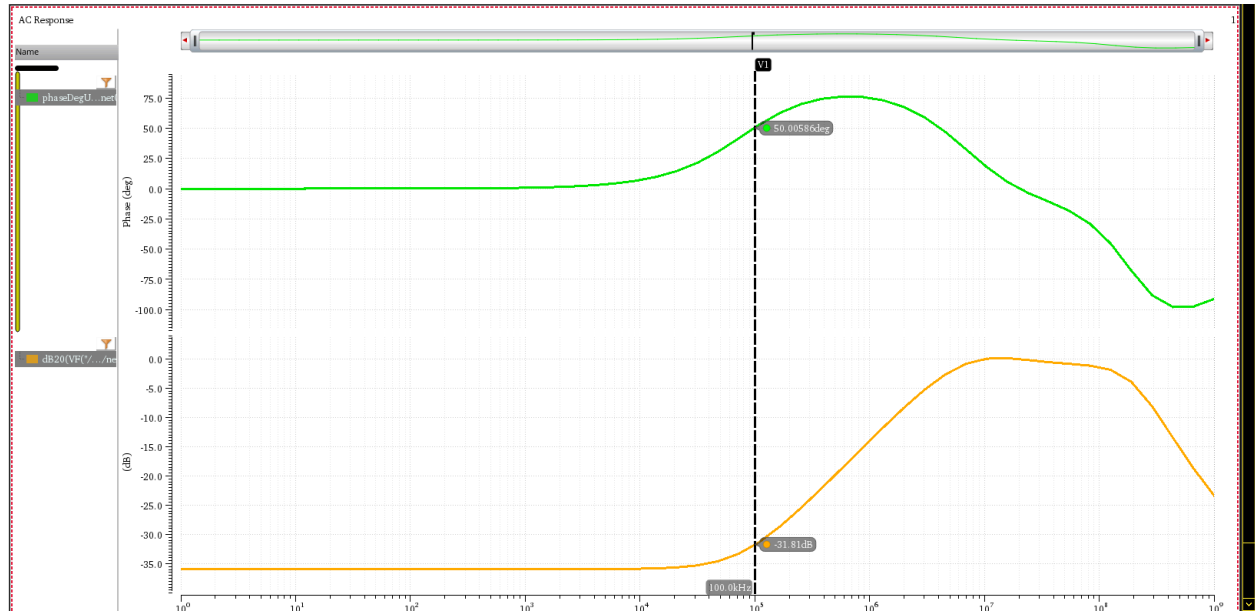
Vin (V)	Vout (V)
1.755	1.49883
1.845	1.50029

$\Rightarrow 2\% \times 1.5 = 0.03 V = 30 mV$

$\Rightarrow \Delta V_{out} = 1.50029 - 1.49883 = 1.46 mV$

$\Rightarrow 1.46 mV \ll 30 mV$

D. PSRR@100KHz(dB): -31.81dB

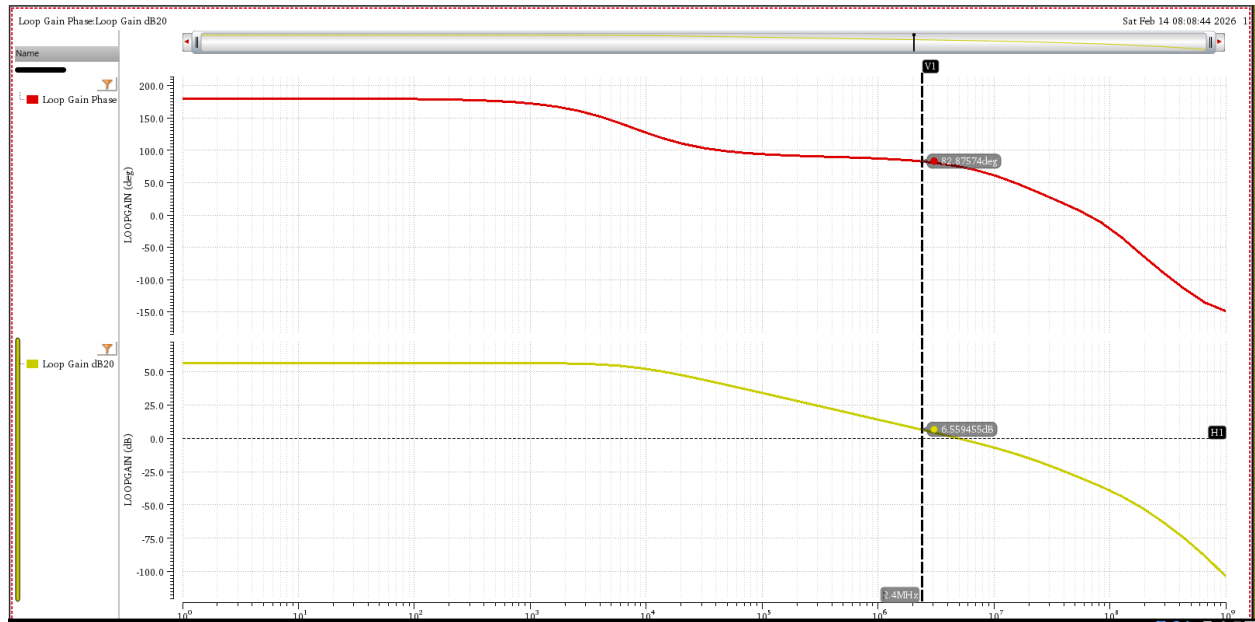


E. Open Loop gain and phase plots

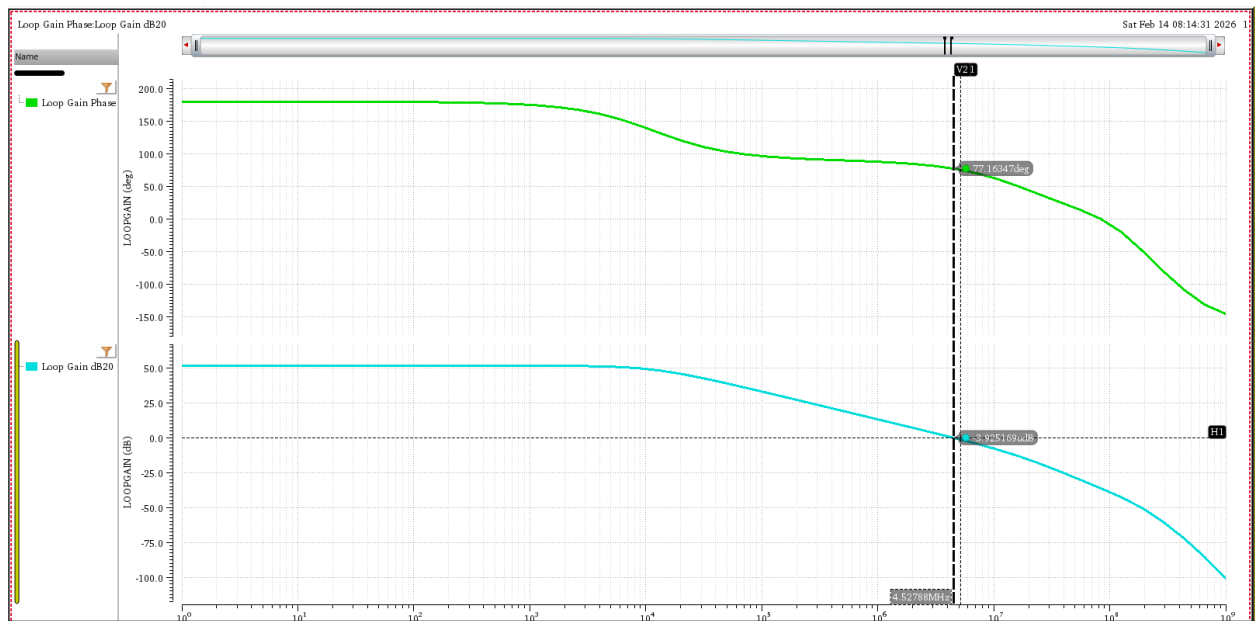
1. Light load



2. Nominal Load

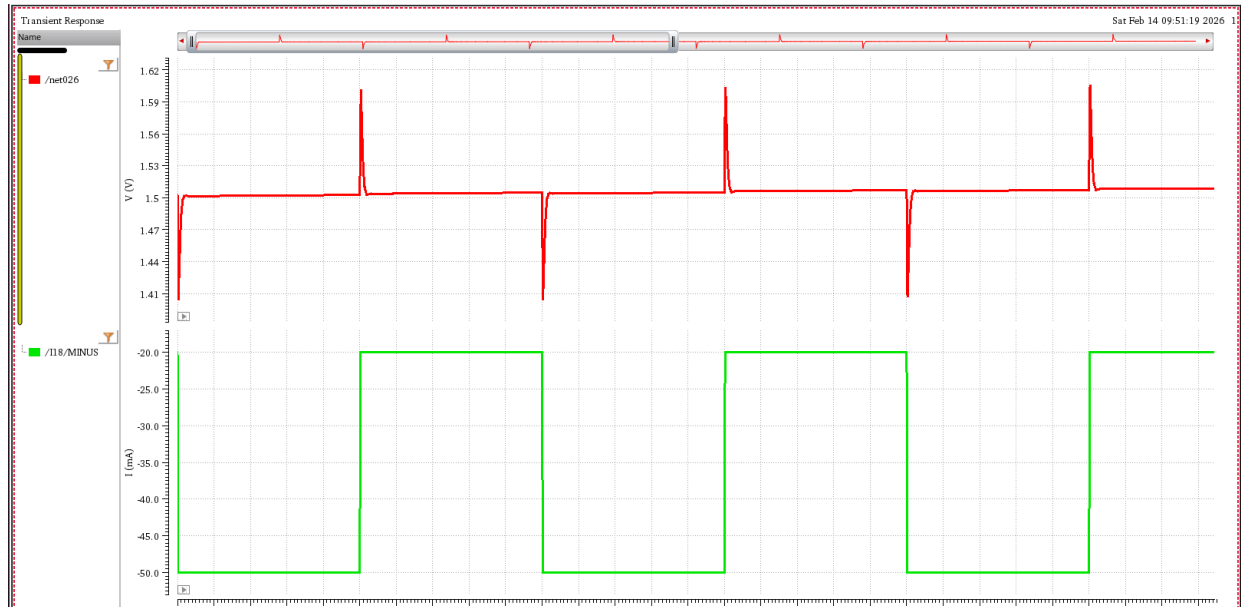


3. Light Load

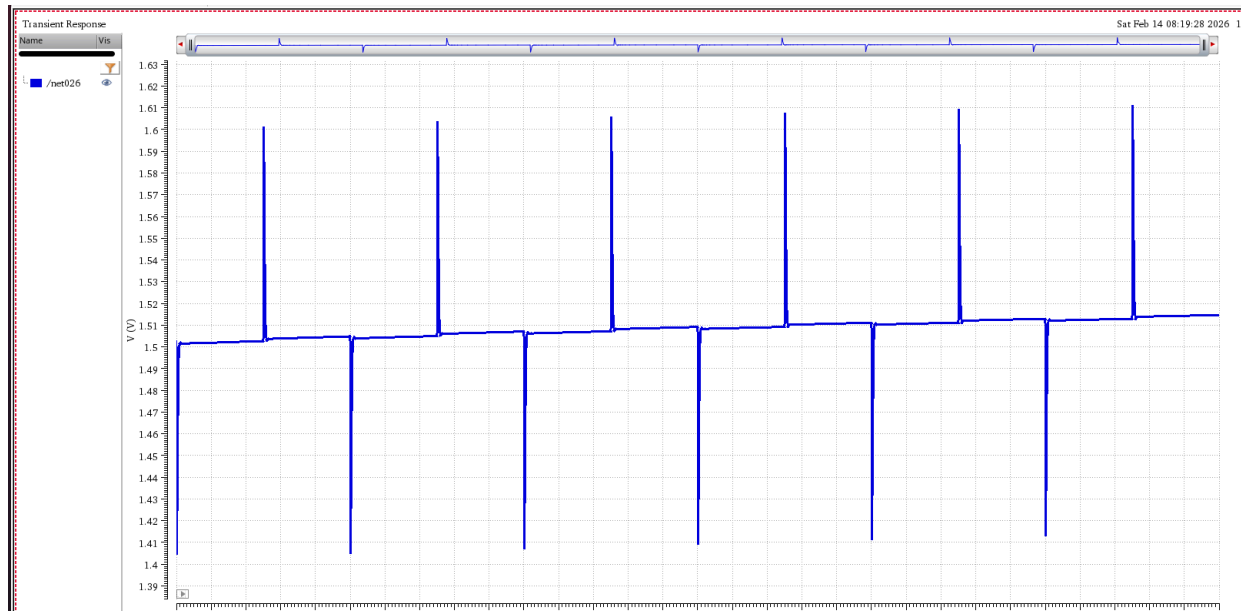


D. Transient Response

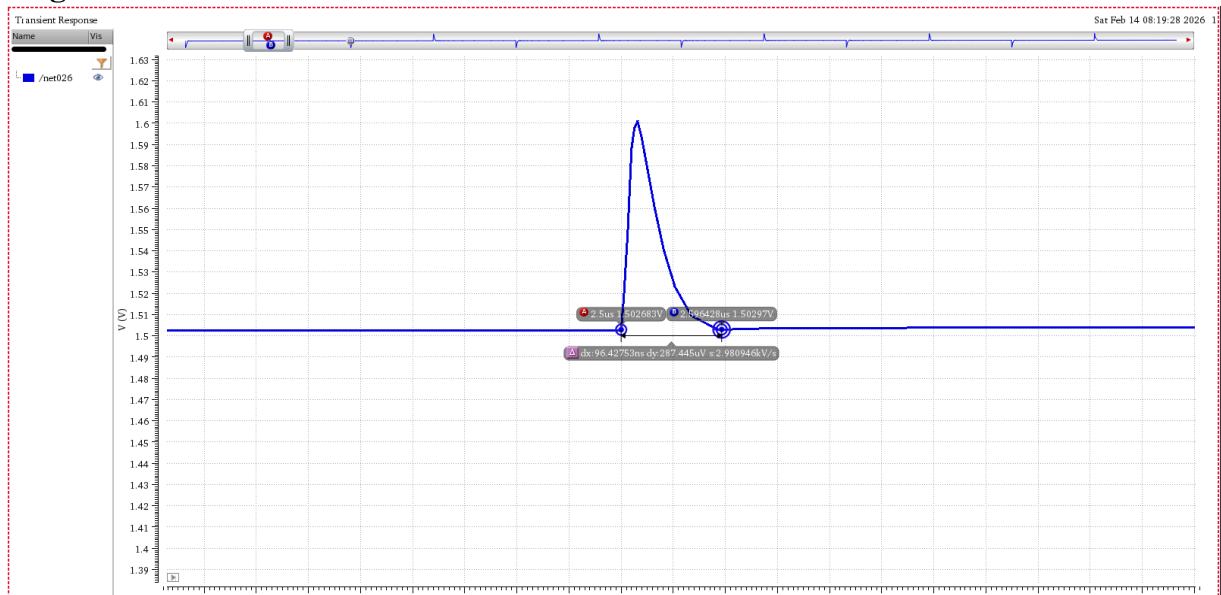
Img 1



Img 2

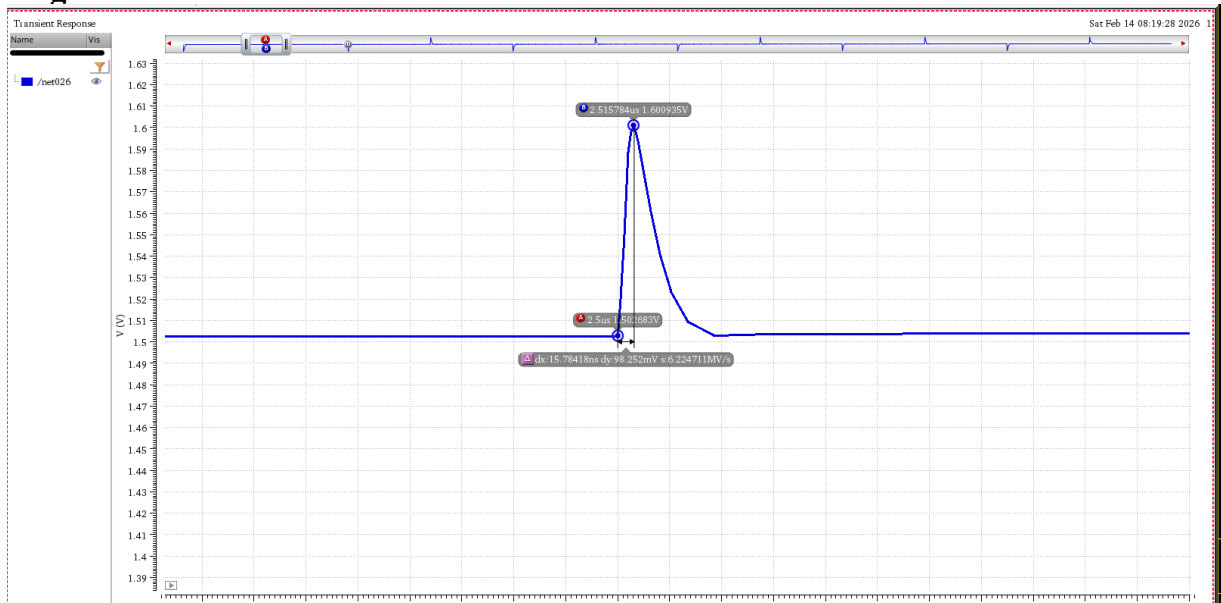


Img 3



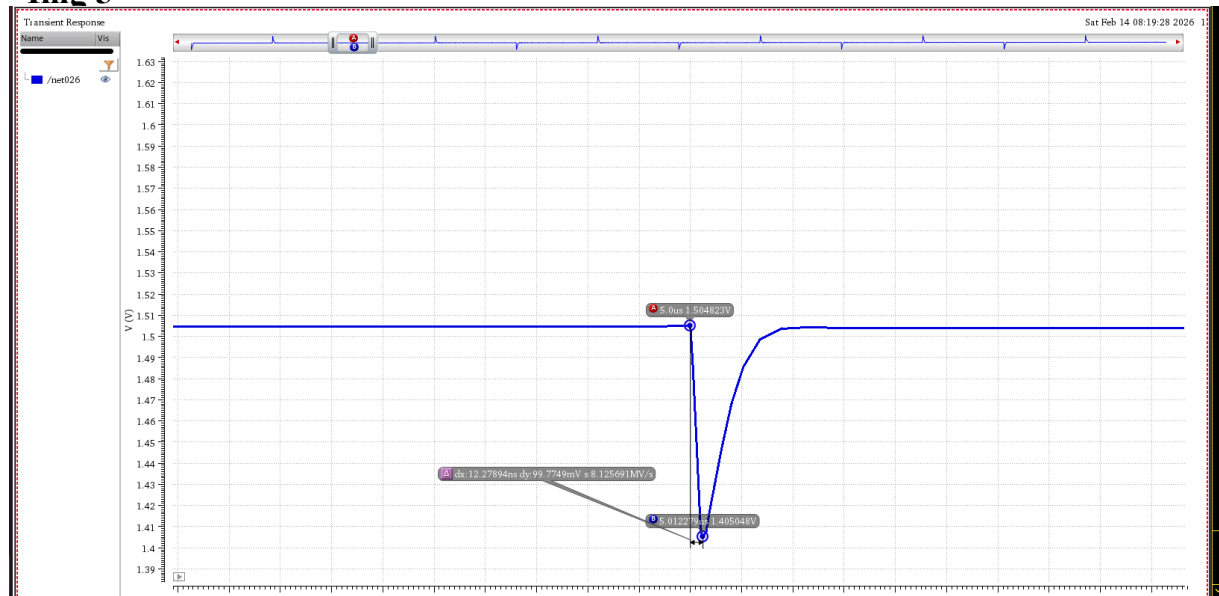
$$\Delta x = 95.42\text{ns}, \Delta y = 287.445\text{uV}$$

Img 4



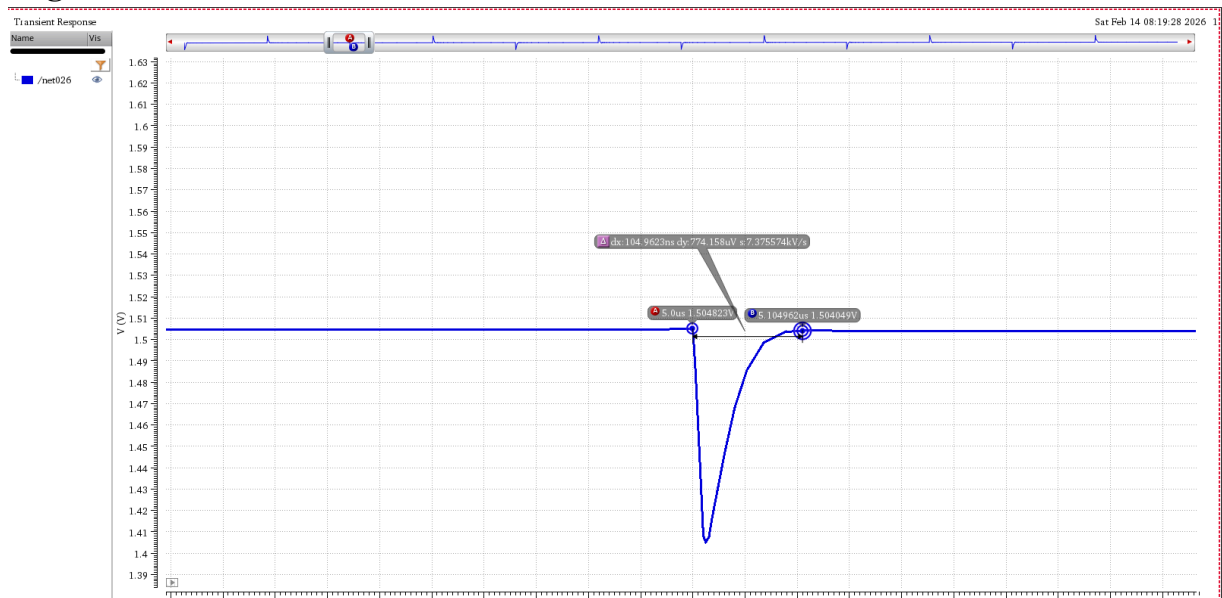
$$\Delta x = 15.781\text{ns}, \Delta y = 98.252\text{mV}$$

Img 5



$$\Delta x = 12.27\text{ns}, \Delta y = 99.77\text{mV}$$

Img 6



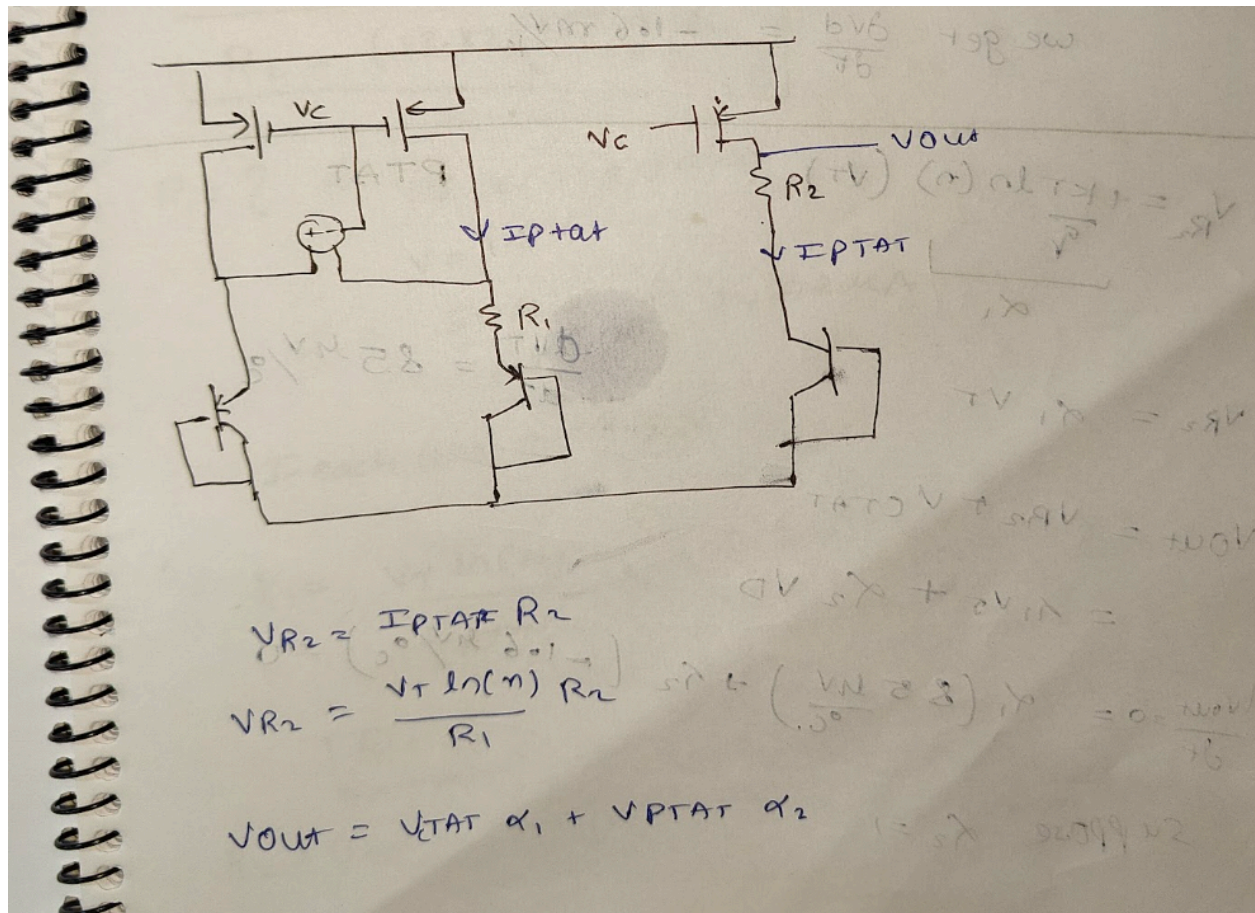
$$\Delta x = 104.96\text{ns}, \Delta y = 774.158\text{uV}$$

E. Efficiency

LOAD	Voltage inp	Voltage out	Current(mA)	Quiscnet current(uA)	Efficiency
Light load	1.8	1.50186	20.0248	151	82.8
Full load	1.8	1.49961	99.9741	151	83.2

Calculations

1. BGR design



$$V_D = V_T \ln \left(\frac{I_0}{I_S} \right)$$

$$\sim \frac{KT}{q}$$

$$I_S \propto M K T n_i^2$$

CTAT

$$\frac{dV_D}{dT}$$

$$I_S \propto b \cdot T^{4+m} e^{-E_g/KT}$$



we will get $\frac{dV_D}{dT} = \frac{dV_T}{dT} \ln \left(\frac{I_0}{I_S} \right)$

$$+ V_T \frac{d}{dT} \left(\ln \left(\frac{I_0}{I_S} \right) \right)$$

$$\Rightarrow \frac{dV_D}{dT} = \frac{V_D - (4+m)V_T - E_g/q}{T}$$

by this equation

$$\text{we get } \frac{\partial V_D}{\partial T} = -1.6 \text{ mV}/^\circ\text{C}$$

$$V_{R2} = \frac{+KT \ln(n)}{q} (V_T)$$

$$\alpha_1$$

$$V_{R2} = \alpha_1 V_T$$

$$V_{out} = V_{R2} + V_{CTAT}$$

$$= \alpha_1 V_T + \alpha_2 V_D$$

$$\frac{dV_{out}}{dT} = 0 = \alpha_1 \left(85 \frac{\mu V}{^\circ C} \right) + \alpha_2 \left(-1.6 \frac{\mu V}{^\circ C} \right)$$

suppose $\alpha_2 = 1$

PTAT

$$\frac{dV_T}{dT} = 85 \mu V/^\circ C$$

$$\alpha_T = \frac{1.6m}{85\mu} = 1.882$$

$$V_{out} = 18.82 V_T + V_D$$

$$= 18.82 (26 \times 10^{-3}) + V_D$$

$$V_{out} = 0.489 + V_D$$

$$\approx 0.7$$

$$V_{out} = 1.2mV$$

$$\alpha_T = 18.82 \Rightarrow \frac{R_2}{R_1} \ln(8)$$

$$R_2 = (18.82 \left(\frac{1}{\ln(8)} \right) R_1$$

$$R_1 = ? \Rightarrow \text{power} = 80 \mu W$$

$$V = 1.8V$$

$$I_T = \frac{50 \mu W}{1.8V} = 27.78 \mu A$$

$$I_{\text{each line}} = 9 \mu A$$

$$R_1 = \frac{V_T \ln(N)}{I_{PTAT}} = \frac{(0.026) \ln(8)}{9 \times 10^{-6}}$$

$$R_1 = 6k$$

$$R_2 = (18.82) \left(\frac{1}{\ln(8)} \right) (6 \times 10^3) = 54.3k$$

$$R_2 = 54.3k$$

2. For calculating the C_c and R_c

we are using compensation cause before compensation we were getting 40 degrees phase margin

$$\omega_{p1} = \frac{\omega_{p2}}{\sqrt{3} A_0}$$

$$\omega_{p2} = 20 \text{ MHz (before compensation)}$$

$$\text{theoretically } \omega_{p2} = \frac{1}{(R_{out})(C_L)} = \frac{1}{(R_L)(C_L)}$$

$$\text{light load } \Rightarrow 75 \Omega = \frac{1}{(75)(500 \times 10^{-12})}$$

$$\omega_{pL} = 26.6 \text{ MHz}$$

In the loop gain plot before compensation we were getting $\omega_{p2} \approx 20 \text{ MHz}$

$$\omega_{p1} = \frac{\omega_{p2}}{\sqrt{3} A_0} = \frac{1}{(R_{out})(C_{net}) 2\pi}$$

$$C_{net} = \frac{\sqrt{3} A_0}{(\omega_{pL})(R_{out}) 2\pi}$$

$$C_{net} = C_{gate} + C_c (1 - (-g_m R_{out}))$$

$$C_{net} = C_{gate} + C_c (g_m R_{out})$$

$\Rightarrow$ while calculating by taking the values from (8.1)
 cadence i got

$$C_{net} = \frac{(1.5 \times 10^{-6})}{(W_{P_1})(R_{out})(2\pi)}$$

$$R_{out} = \left(\frac{1}{g_{ds1}} \right) \parallel \left(\frac{1}{g_{ds2}} \right) = \left(\frac{1}{5.261 \times 10^{-6}} \right) \parallel \left(\frac{1}{4.2237 \times 10^{-6}} \right)$$

$$C_{net} = \frac{7.484 \times 10^{-6}}{(2\pi)(11.54 \times 10^{-6})}$$

 $\Rightarrow C_c = 1.656 \text{ pF}$

$$R_c \gg \frac{1}{g_m}$$

$$R_c = \frac{50}{g_m} = \frac{50}{378.9 \times 10^{-3}} = 131.943$$

Note : In the simulation when i compensated with $C_c = 2\text{pF}$ i was getting PM of around 59 degrees and not able to meet transient specification, so i gradually increased the compensation capacitor value and ended up at 15pF